

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
Enterprise Java

Practical - 1

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1. Implement the following Simple Servlet applications.

a. Create a simple calculator application using servlet.

CODE :-

```
<html>
<head>
<title>Calculator</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initialscale=1.0">
</head>
<body>
<form action="Calculator" method="get" name="frm"> Enter Number
1 :
<input name="num1" type="text"/></br>

Enter Number 2 :
<input name="num2" type="text"/></br>

Operator
<select name="Operator">
<option value="Addition">+</option>
<option value="Subtraction">-</option>
<option value="Multiplication">*</option>
<option value="Division">/</option>
</select>

<input type="reset" value="reset"/>
<input type="Submit" value="submit"/>
</form>
</body>
```

</html>

Calculator logic :-

```
import static java.io.FileDescriptor.out; import
java.io.IOException; import
java.io.PrintWriter; import static
java.lang.System.out; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
```

```
@WebServlet(urlPatterns = {"/Calculator"}) public
class Calculator extends HttpServlet {
```

```
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {        PrintWriter out = response.getWriter();
```

```
        String num1 = request.getParameter("num1");
```

```
        String num2 = request.getParameter("num2");
```

```
        String op = request.getParameter("Operator");
```

```
        if (op.equals("Addition")) {            out.println("Addition is : " +
(Integer.parseInt(num1) + Integer.parseInt(num2)));
```

```
        } else if (op.equals("Subtraction")) {            out.println("Subtraction is : " +
(Integer.parseInt(num1) - Integer.parseInt(num2)));
```

```
        } else if (op.equals("Multiplication")) {            out.println("Multiplication is : " +
(Integer.parseInt(num1) * Integer.parseInt(num2)));
```

```
        } else {
```

```
            out.println("Division is : " + (Integer.parseInt(num1) / Integer.parseInt(num2)));
```

```
        }
```

```
        out.println("Calculator");
```

```
    }
```

```
@Override
```

```
protected void doGet(HttpServletRequest request,
```

```
        HttpServletResponse response) throws ServletException, IOException {
processRequest(request, response);
```

```
    }
```

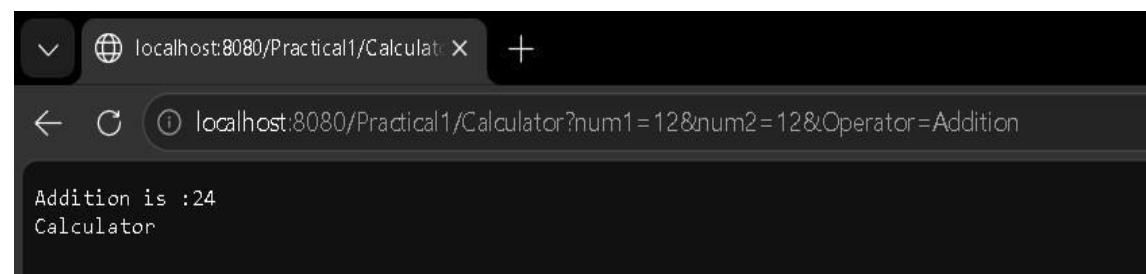
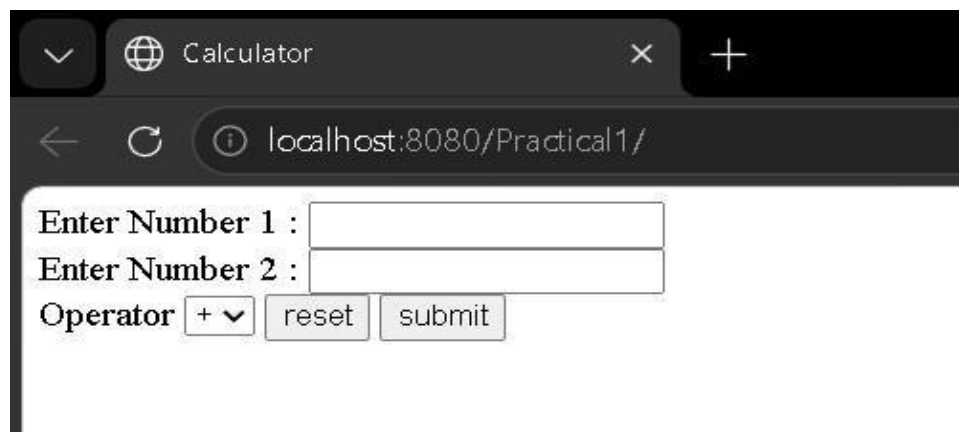
```

@Override
protected void doPost(HttpServletRequest request,
                        HttpServletResponse response) throws ServletException, IOException {
    processRequest(request, response);
}

@Override public String
getServletInfo() {    return
"Short description";
}
}

```

OUTPUT :-



b. Create a servlet for a login page. If the username and password are correct then it says message “Hello <username>” else a message “login failed”.

CODE :-

```

<!DOCTYPE html>
<html>
    <head>

```

```

<title>Login</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
    <form action="login" method="get" name="frm">
Enter username :
        <input name="uname" type="text"/><br><br>

        Enter password :
        <input name="pass" type="password"/><br><br>

        <input name="submit" type="Submit"/>
        <input name="reset" type="Reset"/>
    </form>
</body>
</html>

```

Java Code :-

```

import java.io.IOException; import
java.io.PrintWriter; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;

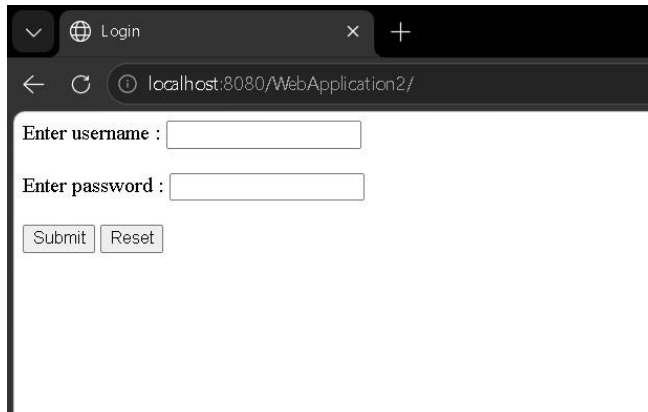
@WebServlet(urlPatterns = {"/login"})
public class login extends HttpServlet {    public void
doGet(HttpServletRequest request,
HttpServletResponse response)        throws
ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {        String
        uname = request.getParameter("uname");
        String pass = request.getParameter("pass");

        if (uname.equals("system") && pass.equals("IT")) {
            out.println("Hello <b>" + uname + "</b> Welcome to Java Servlet");
        } else {
            out.println("Login Failed");
        }
    }
}

```

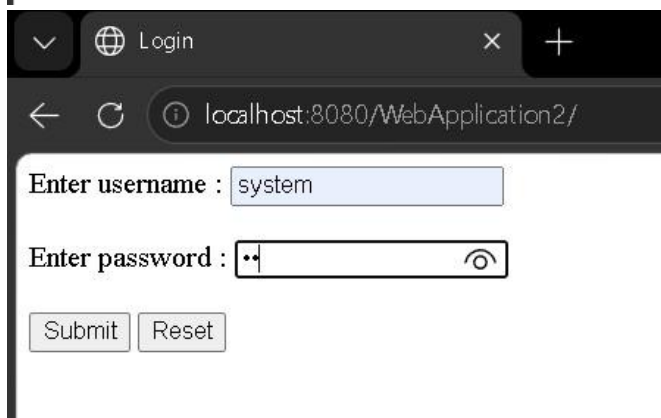
```
}  
}  
}  
}
```

OUTPUT :-



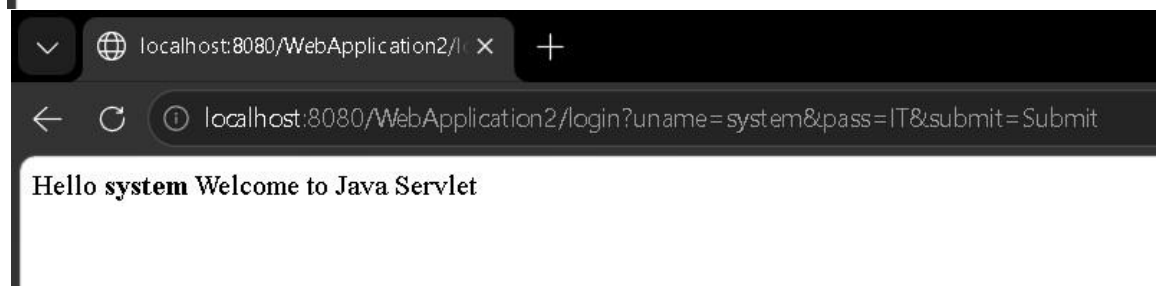
Enter username :

Enter password :



Enter username :

Enter password :



localhost:8080/WebApplication2/login?uname=system&pass=IT&submit=Submit

Hello **system** Welcome to Java Servlet

c. Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

CODE :-

```
<!DOCTYPE html>  
<html>  
<head>
```

```
<title>Practical 1 C</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
<form action="ServReg" method="post">
<table cellpadding="1">
<thead>
<tr>

<th><b>REGISTRATION FORM</b></th>
</tr>
</thead>
<tbody>
<tr>
<td><b>First Name : </b></td>
<td><input type="text" name="fname" value=""></td> </tr>
<tr>
<td><b>Last Name : </b></td>
<td><input type="text" name="sname" value=""></td> </tr>
<tr>
<td><b>Zip Code : </b></td>
<td><input type="text" name="zip" value=""></td> </tr>
<tr>
<td><b>User Name : </b></td>
<td><input type="text" name="uid" value=""></td> </tr>
<tr>
<td><b>Password : </b></td>
<td><input type="password" name="pwd" value=""></td> </tr>
<tr>
<td><b>Confirm Password : </b></td>
<td><input type="password" name="pwd1" value=""></td> </tr>
<tr>
<td><b>Town : </b></td>
<td><input type="text" name="town" value=""></td> </tr>
<tr>
<td><b>Country : </b></td>
<td><input type="text" name="country" value=""></td> </tr>
<tr>
<td><input type="submit" value="Submit"></td> </tr>
</tbody>
</table> </form>
</body>
```

</html>

Java Code :-

```
import java.io.IOException; import
java.io.PrintWriter; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse; import
java.sql.*;
```

```
@WebServlet(urlPatterns = {"/ServReg"}) public
class ServReg extends HttpServlet {
```

```
    @Override    protected void doPost(HttpServletRequest request,
    HttpServletResponse response)        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");        try (PrintWriter out =
    response.getWriter()) {
```

```
        String connectionURL = "jdbc:mysql://localhost:3306/db";
```

```
        Connection connection = null;
```

```
        ResultSet rs;
```

```
        response.setContentType("text/html;charset=UTF-8");
```

```
        String uid = request.getParameter("uid");
```

```
        String fname = request.getParameter("fname");
```

```
        String sname = request.getParameter("sname");
```

```
        String pwd = request.getParameter("pwd");
```

```
        String pwd1 = request.getParameter("pwd1");
```

```
        String town = request.getParameter("town");
```

```
        String country = request.getParameter("country");
```

```
        String zip = request.getParameter("zip");        try {
```

```
            Class.forName("org.gjt.mm.mysql.Driver");
```

```
            connection = DriverManager.getConnection(connectionURL, "root", "root");
```

```
            String sql = "insert into user_register values(?,?,?,?,?,?,?)";
```

```
            PreparedStatement pst;        pst =
```

```
            connection.prepareStatement(sql);        pst.setString(1, uid);
```

```
            pst.setString(2, fname);        pst.setString(3, sname);
```

```
            pst.setString(4, pwd);        pst.setString(5, pwd1);
```

```
            pst.setString(6, town);        pst.setString(7, country);
```

```
            pst.setString(8, zip);        int numRowsChanged =
```

```
            pst.executeUpdate();        out.println("Welcome : ");
```

```
            out.println("'" + fname + "'");        pst.close();
```

```

        } catch (ClassNotFoundException e) {
out.println("Couldnt Load database driver : " + e.getMessage());
        } catch (SQLException e) { out.println("SQL
Exception caught : " + e.getMessage());
        } catch (Exception e) {
out.println(e);        } finally {
try {                if (connection !=
null) {
connection.close();
        }
        } catch (SQLException ignored) {
out.println(ignored);
        }
    }
}
}
}
}
}

```

SQL CODE :-

```

CREATE DATABASE db;
USE db;
CREATE TABLE user_register (
uid CHAR(40), fname
CHAR(40), sname CHAR(40),
pwd CHAR(40), pwd1
CHAR(40), town CHAR(40),
country CHAR(40), zipcode
CHAR(40)
);

```


OUTPUT :-

REGISTRATION FORM

First Name :	Sachin
Last Name :	Mehta
Zip Code :	401105
User Name :	Sach12
Password :	
Confirm Password :	
Town :	mumbai
Country :	india
Submit	

Welcome : 'Sachin'

```
mysql> CREATE DATABASE db;
Query OK, 1 row affected (0.00 sec)

mysql> USE db;
Database changed
mysql> CREATE TABLE user_register (
  -> uid CHAR(40),
  -> fname CHAR(40),
  -> sname CHAR(40),
  -> pwd CHAR(40),
  -> pwd1 CHAR(40),
  -> town CHAR(40),
  -> country CHAR(40),
  -> zipcode CHAR(40)
  -> );
Query OK, 0 rows affected (0.01 sec)
```